CSE546: Reinforcement Learning

Instructor: Alina Vereshchaka
Team 42:
Christian Brazeau, Sean Courtney, Ishaan Malhotra

Problem Statement **3**
Background **4**
Environment **5**
Environments 5
Observations 5
Actions 5
Rewards 5
Chosen Algorithms **6**
Results **7**
Highway DQN Results 7
Test Run 7
Observations **8**
Highway DQN Results 8

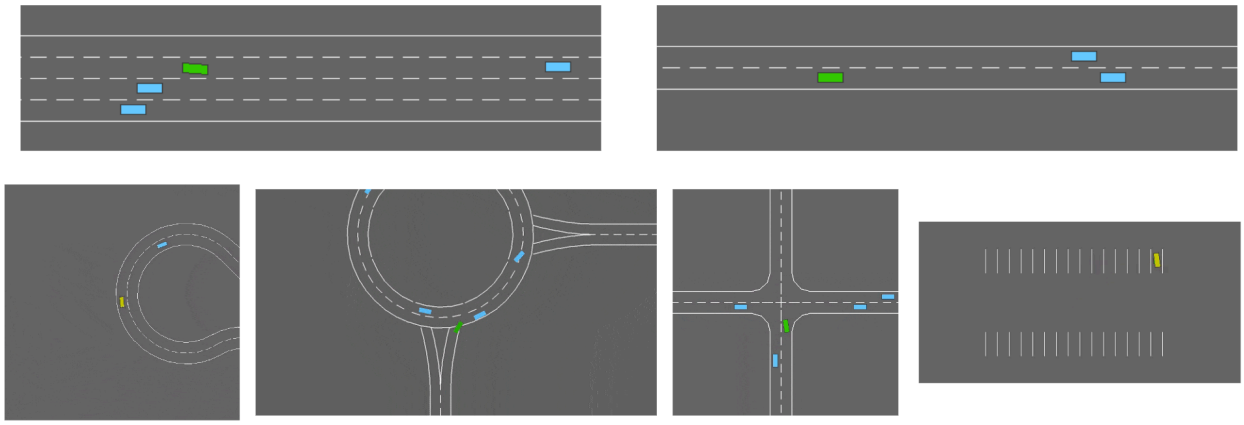
Problem Statement

Autonomous driving has many obstacles and optimizations to overcome in order to be effective in our society. Training using effective simulations for specific use cases such as lane changing, turning in intersections, or parking. Solving these issues will help bring autonomous driving vehicles to the masses in a faster, cheaper, and more effective way. This will lead to less vehicular injuries.

Background

Environment

Environments



Observations

There are several available configurations for the observation space with varying levels of complexity.

Kinematic Observations:

Vehicle	x	y	v_x	v_y
ego-vehicle	5.0	4.0	15.0	0
vehicle 1	-10.0	4.0	12.0	0
vehicle 2	13.0	8.0	13.5	0
…	…	…	…	…
vehicle V	22.2	10.5	18.0	0.5

Occupancy Grid Observations:

The Occupancy Grid Observation is a W x H x F array that represents a grid of shape W x H discretising the space around the ego-vehicle in uniform rectangle cells. Each cell is described by features, listed in the "features" configuration field. The grid size and resolution is defined by the grid_size and grid_steps configuration fields

Time to collision observations:

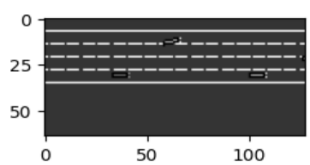
The TimeToCollisionObservation is a V x L x H array, that represents the predicted time-to-collision of observed vehicles on the same road as the ego-vehicle. These predictions are performed for V different values of the ego-vehicle speed, L lanes on the road around the current lane, and represented as one-hot encodings over H discretised time values (bins), with 1s steps
Example (will explain later):

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0

Grayscale image observations:



Actions

Rewards

Chosen Algorithms

Baseline Algorithms

Deep Q-Learning (DQN)

Combines Q-Learning with deep learning to learn a policy for making decisions in an environment. Works only with discrete environments

Dueling Deep Q-Learning (Dueling DQN)

Dueling DQN is an extension on the DQN algorithm that aims to improve efficiency, stability and convergence time by decoupling the estimation of the state value function and action advantages.

Advantage Actor Critic (A2C)

A2C combines the policy gradient method (Actor) and value based method (Critic) to learn a policy for decision making in an environment.

Advanced Algorithms

Deep Deterministic Policy Gradient (DDPG)

Uses Deep Q-Learning and deterministic policy gradient to learn a deterministic policy for continuous action spaces

Asynchronous Advantage Actor Critic (A3C)

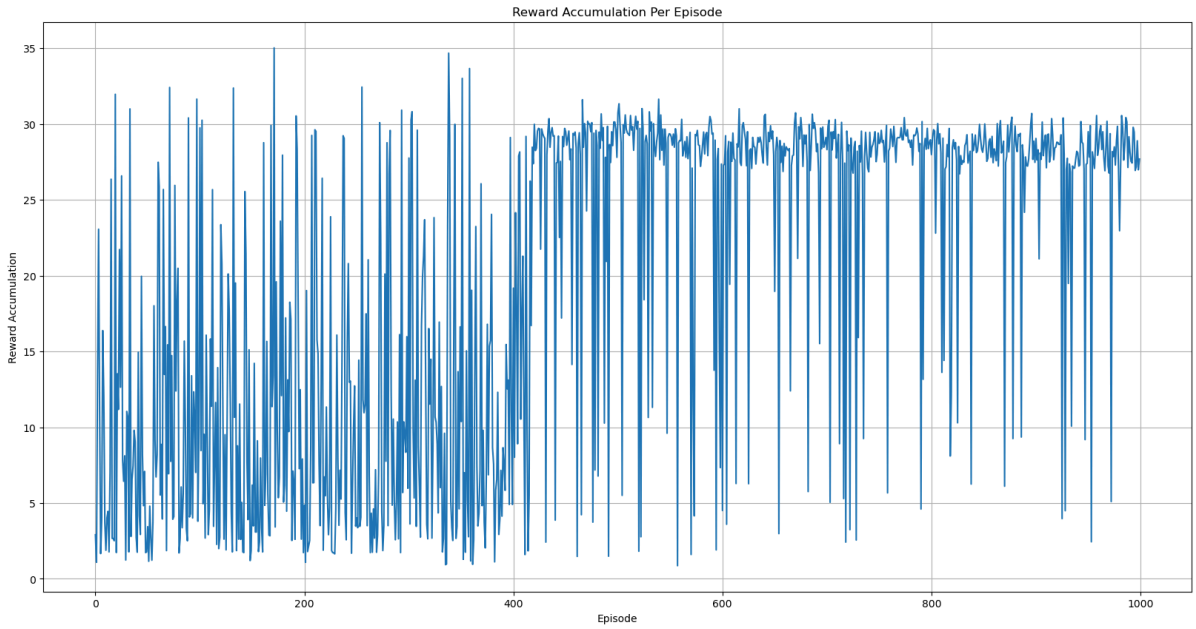
Built on the A2C architecture, where an actor network learns a policy and a critic network evaluates the quality of the actions selected.

Proximal Policy Optimization (PPO)

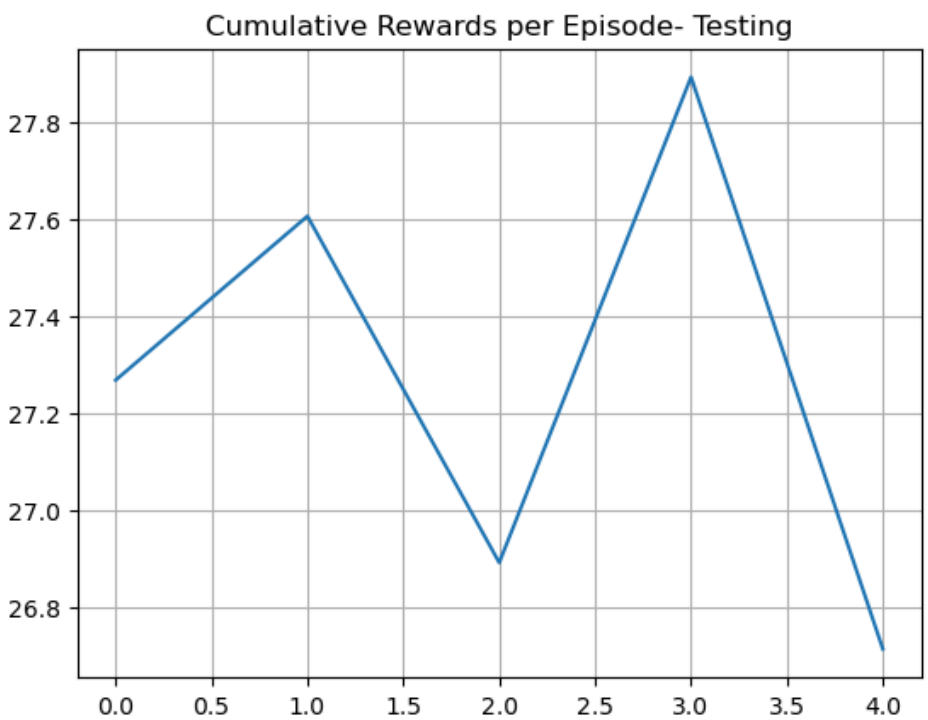
Works to fix the limitations of policy gradient methods (Unstability and slow convergence). This is done by adding a surrogate objective function and clipped policy update mechanism

Results

Highway DQN Results



Test Run



Observations

Highway DQN Results

The DQN ran for 1000 episodes and initially began learning an improved policy nearing the 400 episode mark but then began to stagnate and stop learning until it finished its training cycle. This could be due to the size of the Neural Network being used to train or needing to tune hyper parameters to help the network train in a better fashion.